Wind-wave-tidal-solar complementarity can alleviate the energy crisis on islands: A case of the Zhoushan Archipelago, China

Abstract

Isolated island power systems face significant challenges in integrating intermittent renewable energy resources. This study investigates the complementarity of four renewable resources, including wind, wave, tidal, and solar, in the Zhoushan Archipelago to alleviate local energy shortages. The Entropy Weighting Method (EWM) is applied to optimize resource allocation based on time-series capacity factors. Using ERA5 reanalysis and simulated data, strong natural seasonal complementarities among four resources are identified: (1) high wind and wave output in winter coincides with low solar availability, while abundant summer solar offsets weaker wind and wave resources; (2) tidal resource provides a stable baseline year-round. The results show that the proposed entropy-weighted configuration achieves a more balanced trade-off between energy utilization and output stability than the minimum standard deviation (MSD) approach, which tends to over-concentrate on one resource and sacrifice overall power generation. The integrated system achieves higher capacity factors, lower variability, and reduced curtailment, translating into improved economic performance through higher captured electricity prices. Spatial analysis further identifies two priority development zones with balanced performance and highlights the need for flexible loads in high-resource areas. Overall, this work provides a quantitative foundation for planning multi-energy complementary systems in islands and coastal regions, demonstrating that wind-wave-tidal-solar complementarity can effectively alleviate energy crises and enhance energy resilience.

Keywords: Energy complementarity; Offshore wind resource; Wave resource; Tidal resource; Solar resource; Entropy weighting method; Capacity factor

1 Introduction

As global energy demand continues to grow, traditional fossil fuel–based power systems are increasingly unable to support sustainable development. The accelerating energy transition has therefore placed unprecedented emphasis on exploiting clean and renewable resources [1],[2]. Among them, the oceans, covering more than 70% of the

Earth's surface, hold vast reserves of renewable energy, including offshore wind, wave, tidal, and solar resources (see Fig. 1). Despite their significant potential, marine renewable resources however exhibit strong intermittency and stochastic variability, which fundamentally limits the reliability of single-resource power systems [3].

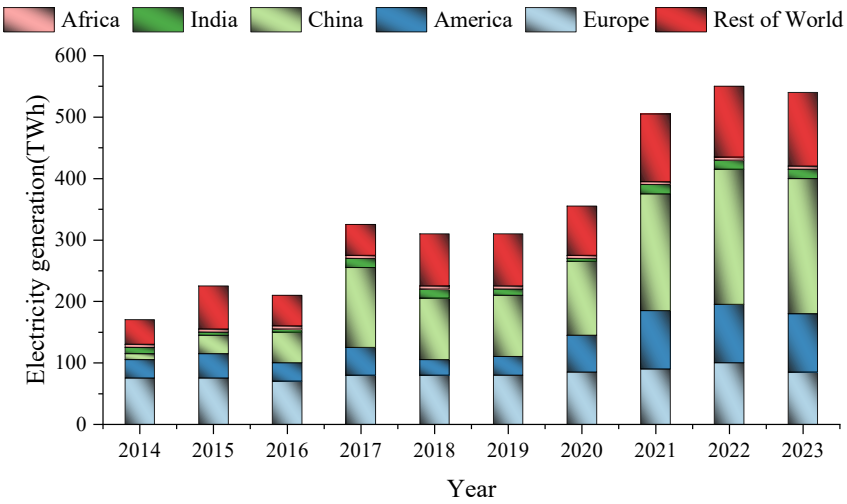


Fig. 1. Renewable energy growth trends (2014-2023)[4]

Ren et al. found that the intermittency of wind power poses obstacles to large-scale grid integration [5]. Petersen et al. reached similar conclusions and further quantified these externalities at the market level through empirical economic studies [6]. Hydropower and photovoltaic power face similar limitations; for instance, Cuartas et al. discovered that extreme drought conditions can severely impact Brazil's energy sector, which relies heavily on hydropower [7].

In response to these challenges, the multi-energy complementary system (MECS) was proposed. Early studies by Al-Ashwal et al. demonstrated that proportionally combining photovoltaic and wind power generation could significantly enhance power reliability, giving rise to the concept of multi-energy complementarity [8].Li et al. and Rozakis et al. introduced dual-energy complementary systems to address the limitations of single-energy power generation[9]. Both designs utilize one or more energy sources to compensate for inherent deficiencies in efficiency or stability [9]. Wang et al. noted that current research has shifted from simply combining multiple energy resources to studying optimal configurations for hybrid systems [10]. For example, Quevedo et al. observed that the annual increase in renewable energy grid integration on islands effectively reduced dependence on fossil fuels. However, they did not analyze power plant regulation performance under different load scenarios (e.g., island peak electricity demand or desalination plant coordinated operation) nor explore optimal coordination

strategies among hydroelectricity, wind power, and diesel generators [11].

Despite these advances, several critical gaps remain. Prasad et al. proposed the utilization of the probability of all wind and solar sources being at either high or low output levels as an indicator of complementarity [12]. Hu et al. argued that while this method provided a precise and intuitive assessment of complementarity, it lacked a mathematical analytical formula. Although it incorporates the coefficient of variation (CV), it fails to account for critical practical planning factors such as the spatiotemporal variability of resources and electricity market mechanisms [8]. Stoutenburg et al. extensively employed capacity factors (CF) to quantify multi-timescale output variability, yet their model entirely disregarded environmental impacts, maritime use conflicts, and planning permit applications [13]. Khan et al. utilized standard deviation (SD) to quantify seasonal fluctuations in wind and solar output, but the Multi-Criteria Analysis (MCA) tool they employed carried significant subjectivity [14].

Meanwhile, the global development of ocean energy has faced significant technical and economic challenges. Most early marine energy pilot projects have suspended or ceased operation due to issues related to device durability, commercial viability, and regulatory constraints (see Fig. 2). These failures highlight the need for more robust scientific assessment tools to evaluate long-term operational feasibility, particularly for offshore multi-energy integration. Such assessments must simultaneously consider energy utilization efficiency, output stability, economic performance, and curtailment risk under realistic operating conditions [15],[16].

Fig. 2. Aguçadoura Wave Farm (left) due to technical reasons, Pelamis Wave Energy (center) due to commercial issues, Roosevelt Island Tidal Energy (right) due to permitting and regulatory challenge.[17]

Despite progress in MECS research, several critical gaps remain. First, most existing studies focus on pairwise resource combinations (e.g., wind+solar or wind+wave), while comprehensive four-source complementarity involving wind, wave, tidal, and solar resources remains understudied, especially in complex archipelagic environments [18]. Second, conventional optimization strategies often rely on historical data to establish fixed weighting schemes, limiting adaptability to dynamic fluctuations

such as seasonal variations in capacity factors [19]. Moreover, these subjective or empirical weight assignments frequently lack a rigorous quantitative foundation, reducing the scientific credibility of the resulting allocation outcomes. Traditional approaches also tend to employ single-indicator optimization or fail to incorporate flexible threshold adaptation mechanism [20][21].

To overcome these limitations, recent studies have introduced objective evaluation methods. Lu et al. utilized the HOMER software to simulate a vast array of feasible configuration schemes and employed the entropy weight method (EWM) to calculate objective weights for multiple indicators, thereby eliminating human subjective evaluation interference [22]; Zhong et al. applied EWM to determine the weights of these conflicting objective functions, subsequently screening the optimal compromise solution from numerous frontier solutions [23]; Shuyi et al. constructed a five-dimensional evaluation index system for MECS [24]; and Lu et al. introduced EWM to establish an evaluation framework for smart energy system development levels, yielding more rational and reliable outcomes [25].

Determining the optimal energy allocation ratio is a critical step toward improving overall power stability and utilization efficiency of MECS. However, as discussed above, conventional optimization strategies face several inherent limitations, including reliance on fixed weighting schemes, subjective empirical assignments, and a lack of flexible threshold adaptation mechanisms. To address these gaps, this study employs EWM to objectively determine capacity allocation ratios for wind, wave, tidal, and solar resources. By integrating rational regional zoning and market-based electricity mechanisms, we develop a multi-energy complementary system tailored to the Zhoushan Archipelago, providing a quantitative foundation for sustainable island energy planning.

2 Materials and Methods

This section details the resource data used in the study, the characteristics of the research area, and the core methodology for optimizing the configuration of the MECS.

2.1 Study area

The study area covers the waters surrounding the Zhoushan Archipelago, extending from approximately 29.4°–31.0° N and 121.5°–123.5° E, as shown in Fig. 3.

Located in the East China Sea, the Zhoushan Archipelago is the largest island group in China, comprising 1,390 islands. Its land area accounts for about 20% of China's total island territory, and hosts a population of roughly one million people [26]. The region features a complex coastal and archipelagic geomorphology with strong and persistent wind resources [27]. It also possesses enormous wave energy potential driven by prevalent swell activity [28]. Moreover, the tidal energy resources in the Zhoushan Archipelago are exceptionally significant: the theoretical average tidal power capacity reaches about 5.6×10^6 kW, accounting for over half of China's total tidal energy potential [29]. Annual sunshine duration exceeds 2,000 hours, providing abundant solar radiation [30]. This combination of natural endowments creates highly favorable conditions for the integrated development of wind, wave, tidal, and solar energy, making the area a promising hub for offshore renewable energy utilization.

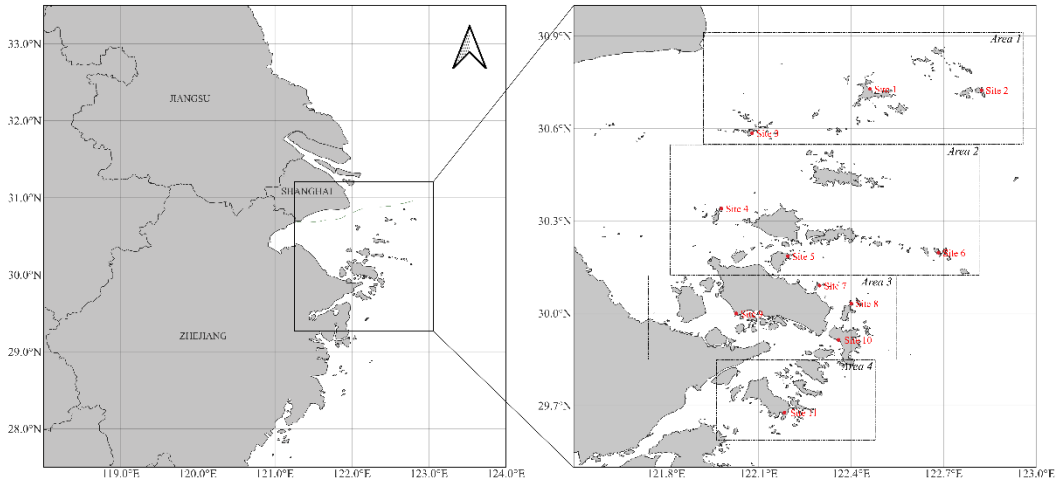


Fig. 3. Location of the study area and the high-potential development zones delineated under the pilot project.

In recent years, the Chinese government has launched a series of multi-energy complementary pilot projects in the Zhoushan Archipelago to explore the coordinated development and efficient utilization of wind, solar, tidal, and even natural gas resources, among others. For instance, Wang et al. conducted equipment intensity and fatigue life assessments for the megawatt-scale pilot project on Xiushand Island (e.g., Site X)[31], concluding that its lifespan could reach 20 years, thereby providing methodological references for reliability verification of other projects [28]. Additionally, the commissioning of the Dongji Island smart microgrid (e.g., Site X) enables the microgrid to operate in synergy with the main grid, providing dual safeguards.

Previous studies have rarely examined detailed regional zoning for the entire Zhoushan Archipelago. Therefore, building upon the achievements and experiences of

these pioneering projects, this research systematically identifies potential areas for multi-energy complementary development within the archipelago. Figure 3 displays the locations of existing projects and four designated development zones, providing a rational and practice-oriented spatial framework for optimizing future offshore multi-energy complementary systems.

To systematically guide regional MECS deployment, the Zhoushan Archipelago is divided into four high-potential development areas (Area 1–4) based primarily on prior pilot project locations. This zoning framework follows administrative boundaries and geographic clustering, providing a clear spatial foundation for subsequent system optimization. The delineated zones enable focused analysis of regional differences in resource availability, infrastructure conditions, and energy demand across the four areas. This facilitates more efficient and cost-effective MECS optimization and allows for the formulation of tailored development strategies for each region based on the findings.

2.2 Data

The baseline resource assessment utilized the ERA5 reanalysis dataset from the European Centre for Medium-Range Weather Forecasts (ECMWF) [32], supplemented by measured wave data from the East China Sea Wave Observation Station, tidal level and current velocity data from the Zhoushan Daishan Tide Gauge Station, and topographic datasets from NOAA and GEBCO [33]. Simulation data were generated using the SWAN and MIKE numerical models [34],[35]. Additionally, technical specifications were collected from domestic and international manufacturers for wind turbines, wave energy converters, tidal energy devices, and photovoltaic modules [36],[37].

2.3 Analytical methods

To quantitatively evaluate the multi-energy complementary system, this study proposes an analytical framework based on the capacity factor time series of wind, wave, tidal, and solar resources. The framework evaluates resource efficiency, temporal variability, and overall system performance in terms of stability, curtailment, and economics. The following subsections describe the main indicators and calculation procedures.

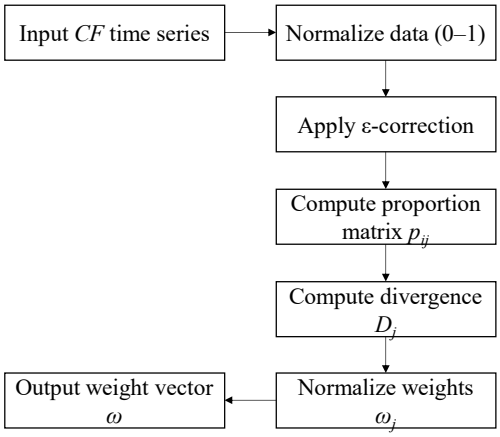
2.3.1 Capacity factor

The capacity factor (CF), defined as the ratio between the actual power output of a device and its rated maximum output over a given period [28], is used to characterize the utilization efficiency of renewable energy resources. As a normalized indicator, CF

reflects both resource availability and the performance of energy conversion devices. In this study, the CF time series of wind, wave, tidal, and solar energy are used as the primary dataset for evaluating the temporal characteristics of different energy sources within the multi-energy complementary system. The CF calculation methods for different energy sources are all derived from the fundamental formula. The fundamental formula and CF calculation steps for the four energy sources are provided in Appendix A.

2.3.2 The entropy weight method

The entropy weight method (EWM) is applied to determine the relative weights of the four energy sources based on the information contained in the hourly CF time series. This method assigns weights according to the variability of each indicator in the dataset, allowing the relative contribution of different energy sources to be quantitatively evaluated. The detailed formulas used for the entropy weight calculation are provided in Appendix A.



Algorithm 1: Entropy Weight Method

Input: CF time-series $\mathbf{X}(m \times n)$

Output: Weight vector ω_j

1: Normalize **X** to obtain **R**
2: Add ε = 1e-6 to avoid log(0)
3: for each indicator j do
4: Compute $p_{ij} = r_{ij} / \mathbf{sum}_i(r_{ij})$
5: Compute entropy $Ej = -k * \Sigma(pij \ln pij)$
6: Compute $D_j = 1 - E_j$
7: end for
8: Compute $\omega_j = D_j / \Sigma D_j$
9: return ω

Fig. 4. Entropy-Based Rights Calculation Process

The complete workflow, including data preprocessing, entropy calculation, and weight normalization, is illustrated in Figure 4, accompanied by pseudocode demonstrating the computational process.

2.3.3 Sensitivity analysis

In addition to determining source weights, the CF time series and the resulting weighted composite output also enable further analyses of system behavior under operational constraints. These include curtailment patterns, economic performance, and capture-price variability. The principle and calculation formula are presented in Appendix C.

2.4 Complementary analysis

The CF calculation methods for different energy sources are all derived from the

fundamental formula. The fundamental formula and CF calculation steps for the four energy sources are provided in Appendix A.Within conventional optimization paradigms, the Minimum Standard Deviation (MSD) approach is commonly employed, in which system stability is maximized by minimizing fluctuations in the total capacity factor (CF_t). The CF_t of the MECS at time t is defined as:

$$CF_t = \sum_{i=1}^{n} \omega_i \cdot CF_{i,t} \tag{1}$$

where $CF_{i,t}$ represents the capacity factor of the i-th energy source at time t, and ω_i is the corresponding entropy weight derived from Section 2.3.

Furthermore, the system's stability is evaluated through the SD of the CF_t, given by:

$$\sigma = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(CF_t - \overline{CF_t})^2} \tag{2}$$

Where N is the number of time intervals, CF_t represents the total capacity factor at time t, and $\overline{CF_t}$ is its mean value. A smaller σ indicates higher temporal stability and stronger complementarity among energy sources.

3 Complementarity Potential Analysis

This section examines the temporal variation and intrinsic complementarity potential among wind, wave, tidal, and solar energy resources in the Zhoushan Archipelago. The analysis focuses on annual and seasonal capacity factor behaviors and their implications for system-level stability in multi-energy configurations.

3.1 Temporal Characteristics of Capacity Factors

The monthly CF variations of the four renewable resources are displayed in Fig. 5. From the figure, all resources demonstrate clear seasonal patterns. Wind energy shows significant seasonal variability, peaking in winter (December–February) under strong monsoon conditions and declining sharply in summer. Wave energy closely aligns with wind patterns, also strengthening in winter. Occasional extreme values appear during typhoon season due to long-period swells. Solar energy exhibits the opposite trend: CF values reach their maximum in summer (June–August) when irradiance is strongest, and they weaken substantially during the winter months. Tidal energy maintains the most stable output among all four sources. Driven by predictable astronomical tidal cycles rather than weather, it shows minimal seasonal variation.

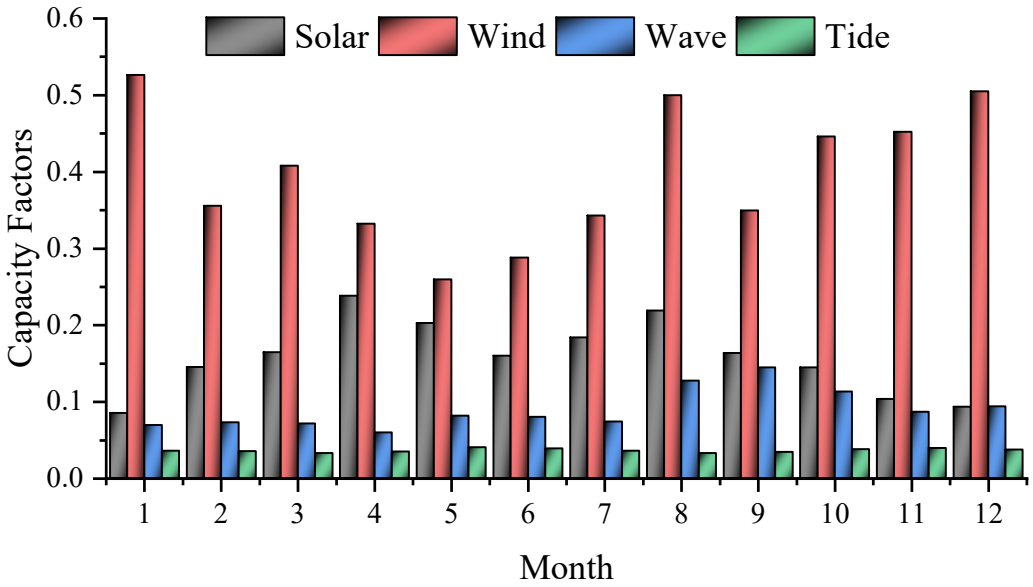


Fig. 5. Monthly CF Change Curves for Each Energy Source.

The quarterly CF distributions (Fig. 6) reinforce these observations: wind and wave energy display high variability, solar energy shows moderate dispersion, and tidal energy remains steadily concentrated with low fluctuations. Taken together, the Zhoushan region demonstrates a characteristic pattern of: wind–wave enhancement during winter, solar dominance during summer, and tidal stability throughout the year. This natural seasonal offset provides a strong basis for multi-energy complementarity.

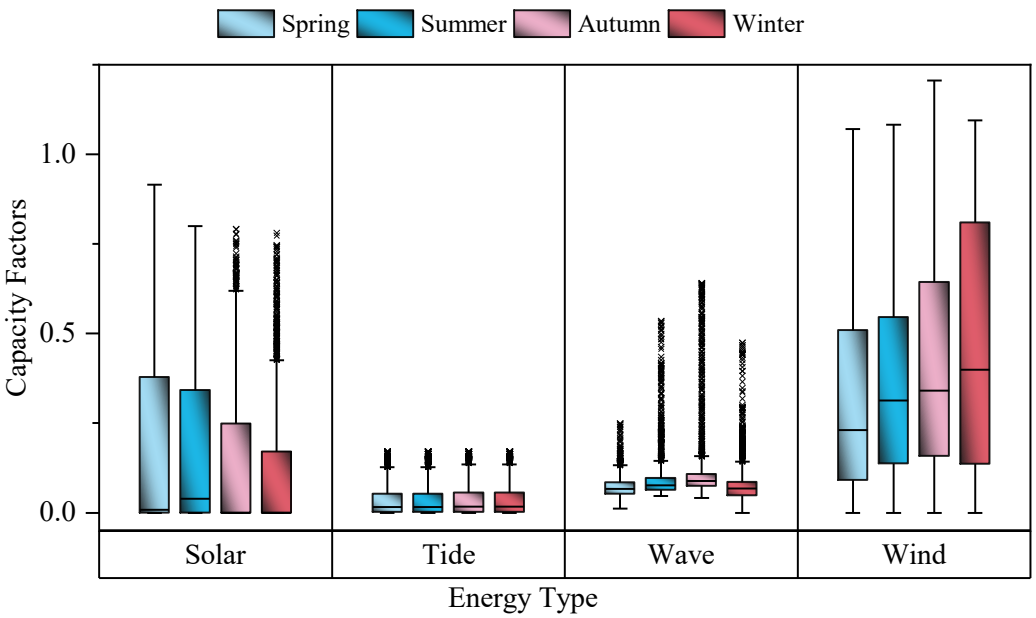


Fig. 6. Quarterly CF Box-and-Line Charts for Each Energy Source.

3.2 Temporal Complementarity Among Energy Sources

This section examines the temporal complementarity among wind, wave, tidal, and solar energy resources by analyzing their capacity factor (CF) variations at different

time scales.

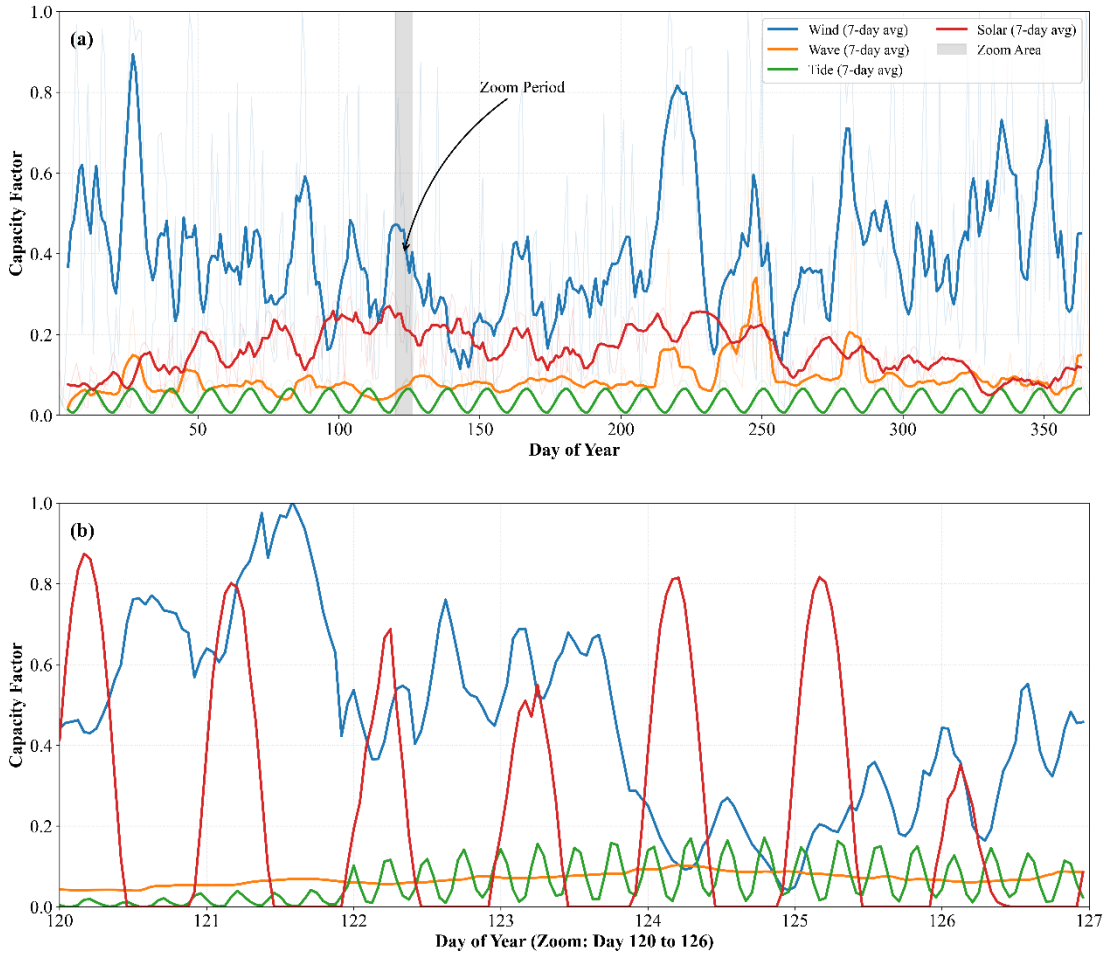


Fig. 7. Temporal complementarity among wind, wave, tidal, and solar energy. The upper **错误!未找到引用源。** (top) illustrates the daily CF variations of wind, wave, tidal, and solar energy throughout 2020. The temporal relationships among the four resources reveal a clear multi-dimensional complementary structure. Wind and wave energy tend to rise and fall together due to their shared meteorological drivers, exhibiting pronounced enhancement during winter monsoons and decreasing significantly during summer. Solar energy shows the opposite behavior, reaching peak CF values during the summer months when wind and wave resources are at their minimum. This seasonal counterbalance between winter-strong wind/wave energy and summer-strong solar energy effectively smooths the long-term output profile of the combined system. Tidal energy, in contrast, remains largely unaffected by atmospheric variability. Its stable, astronomically driven semi-diurnal cycle provides consistent output across all seasons, compensating for the intermittency of the weather-dependent sources and contributing substantially to overall system stability. These complementary temporal dynamics

indicate that the Zhoushan Archipelago naturally supports coordinated multi-energy development, enabling the combined system to maintain higher utilization efficiency and reduced volatility compared with any single resource alone [53].

To further illustrate short-term interactions, Fig. 7 (bottom) shows a zoomed-in view of the hourly CF variations during Days 120–126, extracted from the annual CF time series to provide a representative 168-hour window for detailed analysis. Over this representative 168-hour window, the diurnal nature of solar energy is clearly visible, with high daytime generation and near-zero output at night. Wind and wave energy fluctuate irregularly on hourly timescales yet often co-vary due to their common meteorological forcing. Meanwhile, tidal energy maintains its predictable semi-diurnal periodicity, producing two distinct peaks and troughs each day independent of weather conditions. The interplay of these heterogeneous temporal patterns demonstrates strong short-term complementarity: tidal energy provides a stable baseline during nighttime periods of low solar output, while solar energy offsets tidal troughs during daylight hours. Likewise, the stochastic variations of wind and wave energy are partially absorbed by the regular tidal cycle. This combination of seasonal, diurnal, and semi-diurnal complementarities strengthens the overall robustness of the multi-energy system and enhances its ability to match generation with demand in isolated island grids.

4 Results

4.1 Resource Allocation and Performance

Based on the indicator weights determined by the entropy weighting method, the energy contribution and comprehensive performance of each sub-area were calculated (**Table 1**). The results indicate that the Zhoushan Archipelago's overall resource mix is dominated by tidal energy (31.59%) and solar energy (44.68%), yielding a system-wide composite CF_t of 0.1495 and a standard deviation σ of 0.1114. Significant variations are observed across sub-regions: Area 1 exhibits a very high wave energy share (62.64%) yet demonstrates poor CF_t and stability performance (0.0373 and 0.0357, respectively). Areas 2 and 3 rely primarily on tidal and solar energy, with CF_t values consistently in the range of ~0.145–0.153. Area 4 has a higher proportion of wave energy (34.93%) and demonstrates overall performance in the middle range for both *CFt* and σ.

Table 1

Calculated CF_t and σ of the MECS based on the EWM.

Area	Wind	Wave	Tide	Solar	CF_t	σ

Total	14.84%	8.90%	31.59%	44.68%	0.1495	0.1114
Area 1	9.10%	62.64%	14.56%	13.70%	0.0373	0.0357
Area 2	16.06%	8.82%	34.44%	40.68%	0.1454	0.1085
Area 3	15.90%	11.04%	33.18%	39.88%	0.1526	0.1097
Area 4	12.00%	34.93%	24.59%	28.49%	0.1188	0.0850

Further analysis combining power output and economic indicators (**Table 2**) reveals additional insights. Area 4 exhibits the highest installed capacity and total annual generation, yet it also faces higher curtailment risk and lower captured electricity prices. This indicates that resource development in Area 4 would require supporting measures such as energy storage or flexible loads (e.g., hydrogen production, desalination) to absorb its fluctuating output [51]. Areas 2 and 3 achieve a favorable balance between output stability and generation efficiency, making them suitable as priority zones for grid-connected development. In contrast, Area 1—while having the lowest variability—delivers very low overall output, rendering it unsuitable as a primary development zone.

Table 2

Power generation efficiency, electricity generation volume, and average power output by region.

Area	P_{Rated}(kW)	E_{Total}(MWh)	P_{Avg}(kW)
Total	1218.7	1600.3	182.19
Area 1	4105.1	1345.1	153.13
Area 2	1271	1623.3	184.8
Area 3	1391.8	1865.4	212.37
Area 4	2617.6	2732.4	311.07

As shown in **Table 1**, the Zhoushan Archipelago's optimal configuration (via EWM) primarily emphasizes solar and tidal energy, with wind and wave energy constituting smaller shares. Although solar energy exhibits the lowest CF during winter, its inverse seasonal correlation with wind energy's variability confirms solar's value as a key component of the MECS by providing a counterbalancing effect. Concurrently, tidal energy's consistently high CF throughout the year and very low variability lead to a low entropy (high weight) assignment, underscoring its crucial role for system stability. Wave energy's CF follows a seasonal pattern similar to wind—higher in autumn and winter, lower in summer—and its significant variability results in the lowest weight among the four sources.

Different optimization strategies may lead to distinct energy allocation outcomes. Therefore, the entropy weight method (EWM) was compared with the minimum

standard deviation (MSD) approach to examine their differences in system performance 错误!未找到引用源。错误!未找到引用源。. The results of calculating the weighting ratios for various energy sources in the region using the MSD method are shown in Table 3.

Table 3

Calculated CF_t and σ of the MECS based on the MSD method.

Area	Wind	Wave	Tide	Solar	CF_t	σ
Area 1	0.00171190%	99.90%	0.07%	0.03%	0.0006	0.0006
Area 2	0.00000083%	79.96%	18.13%	1.91%	0.0479	0.0327
Area 3	0.00000002%	84.20%	13.17%	2.63%	0.0464	0.0370
Area 4	0.00000603%	97.71%	1.31%	0.98%	0.0181	0.0177

The MSD method, in its pursuit of the lowest variability, yields extremely concentrated weighting distributions. For instance, in Area 1 nearly all weight (~99.9%) is allocated to wave energy, while wind, solar, and tidal energy together account for less than 0.1%. Similarly, Area 4's mix is over 97% wave energy. In contrast, the entropy-weighted method produces a much more balanced allocation, even in areas where one resource dominates. Notably, in the entropy scheme (Table 1) Areas 2 and 3 have wave energy as a major component but still retain significant contributions from other sources. This aligns with the natural complementary characteristics of the Zhoushan Archipelago's resource endowment.

Calculations of power generation performance under the MSD method are shown in **Table 4**. Although the instantaneous power P (kW) in each area can reach relatively high levels—for example, peak outputs of approximately 5,894 kW in Area 1 and 5,769 kW in Area 4, demonstrating strong short-term resource potential—further analysis of annual energy output E(MWh) and average power $PAvg$ (kW) reveals that the actual available energy is significantly lower. Taking Area 1 as an example: despite a peak approaching 6 MW, its annual energy output is only 8.426 MWh, corresponding to an average power of less than 1 kW. Other areas similarly exhibit the characteristic of "strong peaks but weak averages." This indicates that these MSD-weighted configurations suffer from high fluctuations and low stability, making them unsuitable for sustained power generation or long-term deployment. Therefore, employing the entropy weighting method for resource allocation can substantially reduce volatility while maintaining usable output.

Table 4

Calculated CF_t and σ of the MECS based on the MSD.

Area	P(kW)	E(MWh)	P_Avg(kW)
Area 1	5894.4	8.426	0.95924
Area 2	4772.3	2392.6	272.38
Area 3	5007.6	2236.6	254.62
Area 4	5768.9	605.15	68.892

Regarding the trade-off between stability and utilization, the MSD method does achieve the lowest σ in Area 1 and Area 4, values significantly lower than those under the entropy-weighted method—reflecting its single-objective optimization for stability. However, while attaining these very low variability levels, the MSD method incurs a drastic reduction in CF_t. For example, Area 1's CF_t under MSD is only 0.0006 (**Table 3**), far below the optimized result achieved with EWM (0.0373). This confirms our earlier observation: solely targeting minimum σ can sacrifice the system's average output and overall energy utilization due to overly concentrated resource allocation.

Based on the objective weights derived from the entropy method, a practical configuration scheme for the Zhoushan MECS would use solar and tidal energy as the primary sources, with wind and wave energy as supplementary sources, to achieve an optimal balance between utilization rate and output stability. This objective weighting approach effectively avoids the subjectivity in weight determination that is inherent in traditional configuration methods.

Evaluation results for typical pilot sites S1–S4 (**Table 5**) further validate the presence of regional differences. The composite *CFt* values at each site range from about 0.09 to 0.11, with σ values below the overall regional average. Sites S1 and S4 benefit significantly from high tidal energy contributions; site S3 is dominated by wave energy; and site S2 features a higher proportion of solar power. The comparison indicates that the entropy-weighted configuration provides a more balanced allocation among energy sources, which is consistent with the complementary characteristics observed in the Zhoushan Archipelago.

Table 5

Calculation results of typical pilot projects based on the EWM.

Area	Wind	Wave	Tide	Solar	CF_t	σ
S1	13.20%	0.00%	56.70%	30.11%	0.1035	0.0821
S2	12.06%	13.17%	44.70%	30.07%	0.1115	0.0837
S3	9.25%	49.37%	19.66%	21.72%	0.0882	0.0636
S4	9.81%	4.95%	58.03%	27.21%	0.0997	0.0724

4.2 Sensitivity Analysis

To further assess economic impacts, a sensitivity analysis was conducted on

curtailed electricity and captured electricity prices (**Table 6**). The results indicate that Area 4 exhibits the highest curtailed energy volume and rate (587.93 MWh wasted, 21.52% curtailment) and the lowest captured electricity price (0.3186 yuan/kWh), reflecting significant market value erosion due to frequent oversupply. Area 2, on the other hand, shows the lowest curtailment rate (6.08%) and the highest captured price (0.3733 yuan/kWh), highlighting its excellent load matching characteristics. After optimization using the entropy weight method, the system's overall price discount rate (relative reduction in market value due to timing of generation) decreased from about 20.35% to 17.77%, significantly improving economic returns. This indicates that multi-energy complementary configurations not only enhance operational stability but also mitigate the price erosion effect of renewable energy surplus in the market.

Table 6

Wasted electricity volume, wasted electricity rate, actual annual total power generation, capture cost.

Area	$E_{Curtailed}$ (MWh)	Curtailment Rate	E_{Sold} (MWh)	Pcap(Yuan/kWh)
Total	96.88	6.05%	1503.5	0.3729
Area 1	95.15	7.07%	1250.0	0.3693
Area 2	98.62	6.08%	1524.6	0.3733
Area 3	161.90	8.68%	1703.5	0.3650
Area 4	587.93	21.52%	2144.5	0.3186

Fig. 8 displays the power generation curves and curtailed electricity volumes for the Zhoushan Archipelago over the first 168 hours of 2020. In this simulation, the island's daily load profile was represented by a sine function to approximate typical morning and evening peak demand periods, closely mirroring real-life scenarios. The figure reveals that the overall power generation curve exhibits distinctive offshore renewable characteristics: high volatility and pronounced intermittency. Regions 1 and 4, dominated by wind and wave energy, show large fluctuations that frequently exceed the local load capacity. These meteorologically driven surges result in structurally high curtailment levels, indicating a strong need for flexible resources or additional storage capacity. This pattern is representative of many wind/wave-dominated island systems globally and is an important consideration for resource planning.

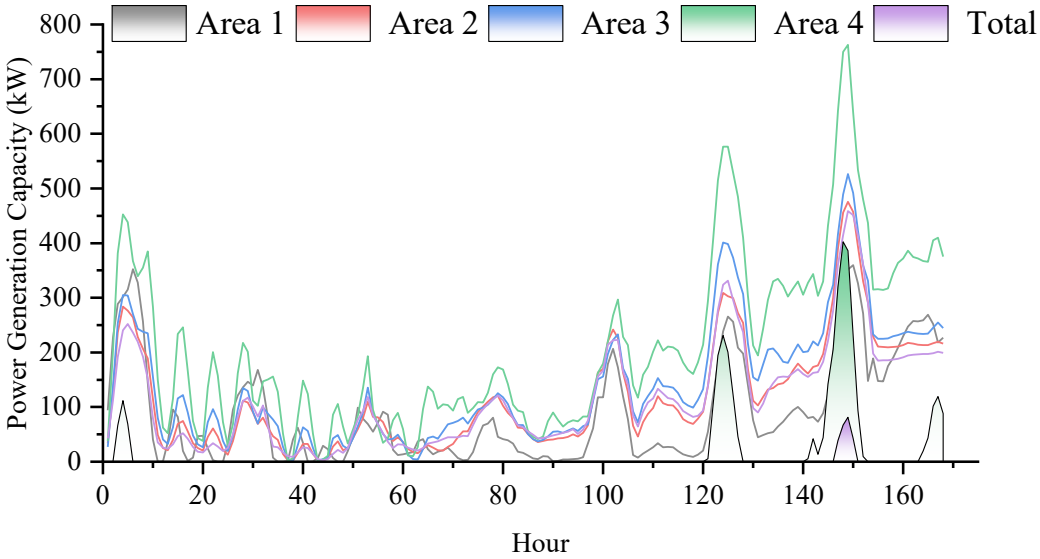


Fig. 8. Power generation curves and curtailed electricity over the first 168 hours of 2020 in the Zhoushan Archipelago.

Conversely, Regions 2 and 3 maintain much steadier generation, with only mild deviations from the load curve. Their smoother profiles can be attributed to the persistent, weather-independent nature of tidal energy, complemented by the predictable diurnal pattern of solar irradiance. The reduced curtailment in these regions directly supports their identification as priority zones for grid-connected development, as they naturally exhibit higher compatibility with typical island load patterns.

A key insight revealed by the total system curve is the strong smoothing effect achieved through spatial and multi-source aggregation. When the outputs of the four regions are combined, the resulting fluctuations are significantly dampened and the occurrence of extreme peaks is substantially mitigated. This temporal offsetting among regions enables the integrated system to follow the load more effectively and reduces dependence on high-cost balancing resources. Such behavior is consistent with the theoretical understanding of multi-energy complementarity and reinforces the value of coordinated planning across the archipelago rather than isolated regional development.

In summary, the Zhoushan Archipelago's renewable resources exhibit characteristic negative correlations and complementary behavior on a seasonal scale. The stable output of tidal energy and the winter enhancement of wave energy jointly reinforce the overall synergy of the system. The entropy weight method effectively balances utilization rate and stability in allocating resource weights, demonstrating greater robustness and practical value in engineering applications compared to the MSD method.

5 Discussion

5.1 Implications of the Entropy Weight–Based Configuration

This study systematically analyzes the spatiotemporal complementary relationships among wind, wave, tidal, and solar energy in the Zhoushan Archipelago, and explores how this multi-energy synergy can enhance system stability, reduce energy storage requirements, and improve economic performance. As shown in Fig. 9, the four energy sources exhibit distinct phase differences on a seasonal scale: in winter, influenced by monsoons, wind and wave energy significantly increase, while in summer, solar radiation intensity is highest, and solar power generation capacity reaches its peak. In contrast, tidal energy is governed by astronomical tidal cycles and maintains a stable output throughout the year.

When the four energy sources are developed in combination, the system's composite capacity factor (CFt) curve is noticeably smoother than that of a single energy source (Fig. 9) 错误!未找到引用源。.. The reduction in monthly peak-to-valley differences and overall fluctuation amplitude indicates a significant decrease in the demand for long-duration energy storage systems. In isolated island power systems, energy storage capacity is typically used primarily to balance seasonal energy fluctuations, whereas multi-energy complementarity can, to some extent, substitute for the regulation functions performed by energy storage systems. This result demonstrates that the complementary relationship among natural resources can, to a certain extent, reduce reliance on large-scale energy storage facilities, thereby improving the overall operational efficiency of the system 错误!未找到引用源。.

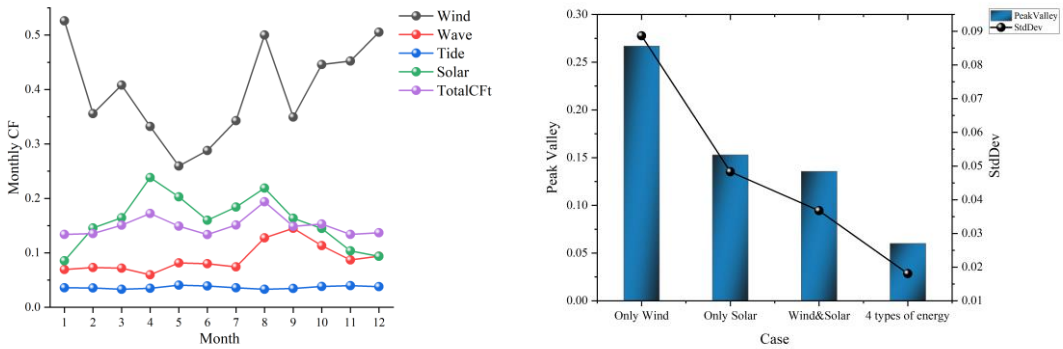


Fig. 9. Seasonal variations of wind, wave, tidal, and solar capacity factors and the resulting smoothed composite CFt curve.

System performance is also significantly influenced by the method used to allocate energy weights. A comparison of the results from the Entropy Weighting Method

(EWM) and the Minimum Standard Deviation Method (MSD) reveals distinct differences in system performance between the two approaches. As shown in Fig. 10, under the EWM configuration, the CFt values for each region remain within a reasonable operating range, and annual power generation remains at a high level. In contrast, the MSD method, in its pursuit of minimal volatility, tends to concentrate weights extremely heavily on a single low-volatility resource (such as wave or tidal energy), resulting in a significant decrease in the system's average output.

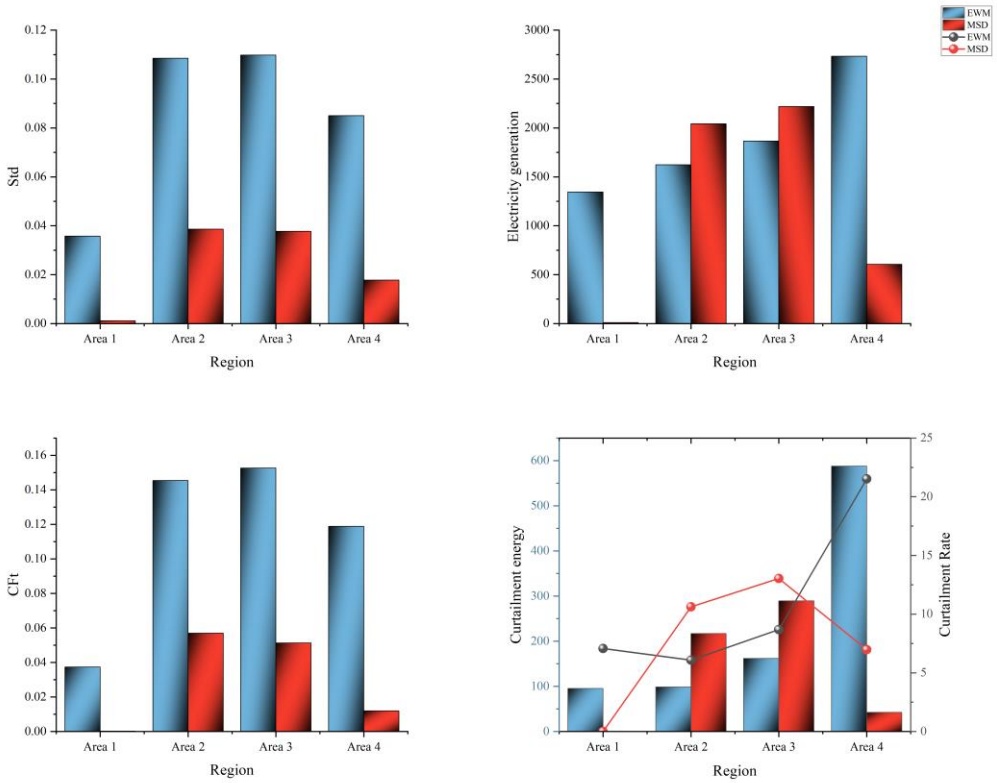


Fig. 10. Comparison of system performance under the EWM and MSD weighting schemes, including CFt, variability, and curtailment behavior.

In some regions, the CFt obtained using the MSD method was even close to zero, indicating that optimization aimed solely at reducing volatility is not suitable for this project's planning. The entropy-weighted method allocates weights based on the information content of the capacity factor time series, enabling the control of volatility levels while maintaining the system's available generation capacity. This achieves a more reasonable balance between energy utilization and system stability, making it more suitable for the actual deployment of this project.

Furthermore, the configuration results across different regions reveal distinct spatial variations. Regions 2 and 3 exhibit relatively balanced characteristics in terms

of CFt, volatility, and power utilization efficiency, while also featuring low curtailment rates and a high degree of alignment with local load curves. Therefore, these two regions are suitable as priority areas for grid-connected offshore multi-energy development. In contrast, although Region 4 possesses significant resource potential, its high output fluctuations result in a high curtailment rate and relatively low economic returns. Consequently, its development requires the integration of flexible loads—such as hydrogen production, desalination, or cold-chain energy storage—to absorb excess electricity.

Region 1, due to the disproportionately high share of wave energy, has a relatively small overall output scale and is therefore more suitable for demonstration or technology validation projects rather than large-scale development.

5.2 Characteristics of curtailed electricity and their economic impact

In addition to system stability, multi-energy configuration also has a significant impact on curtailment behavior and economic performance. As shown in the results of Table 6, there are marked differences among regions in terms of curtailed power, curtailment rate, and captured electricity price.

Among these, Region 4 had the highest curtailed power, reaching 587.93 MWh, with a curtailment rate of 21.52%; at the same time, it had the lowest captured electricity price. This indicates that under high-resource conditions, this region is prone to excess generation, resulting in a large amount of electricity that cannot be effectively utilized, thereby reducing overall economic returns. In contrast, Region 2 has the lowest curtailment rate, at only 6.08%, and the highest captured electricity price, suggesting that its generation time distribution aligns better with electricity demand.

Further analysis of the generation curves reveals significant differences in the fluctuation characteristics of power generation across different regions. As shown in Fig. 8, during the first 168 hours of the simulation, the generation curves for Region 1 and Region 4 exhibited significant fluctuations, with power generation frequently exceeding load demand, resulting in high levels of curtailed power. This phenomenon is primarily due to the fact that wind and wave energy are driven by meteorological conditions, leading to high uncertainty and short-term volatility in their output.

In contrast, the power generation curves for Zones 2 and 3 are smoother and better aligned with the load curve. This is primarily because tidal energy provides a stable, periodic output, while solar energy exhibits regular diurnal variations; combining the two can smooth the system's output to some extent, thereby reducing the risk of

curtailed generation.

Furthermore, when the power generation outputs from the four regions are combined, the overall fluctuation of the system is significantly reduced. As shown by the total power generation curve in Fig. 8, the joint development of multiple regions and multiple energy sources can effectively mitigate extreme output fluctuations in individual regions, reduce the frequency of peak power occurrences, and thereby enhance the system's adaptability to load variations.

From an economic perspective, a complementary multi-energy configuration not only increases total system generation but also improves the capture of electricity prices during the power sales phase. This means that more electricity can be sold during periods when market prices are higher, thereby increasing overall revenue. In contrast, if the energy mix is overly concentrated on a single resource type, it is prone to generating large amounts of power during low-price periods, leading to increased price erosion.

6 Conclusion

This study develops an objective entropy-weighted framework to quantify the complementarity of wind, wave, tidal, and solar resources in the Zhoushan Archipelago and evaluates the performance of multi-energy configurations under realistic operational constraints. By integrating long-term ERA5 time-series data with a data-driven weighting method, the analysis reveals a strong seasonal inverse correlation between wind and solar resources, reinforced by winter enhancement of wind–wave energy and the year-round stability of tidal energy. This inherent four-source synergy significantly reduces seasonal fluctuations in the composite capacity factor, thereby lowering the requirement for large-scale energy storage in island power systems.

Compared with the minimum-standard-deviation approach, the entropy-weighted configuration achieves a more effective balance between usable output and temporal stability. The method avoids excessive weight concentration on low-output resources and maintains a meaningful average generation while keeping system variability within operationally acceptable limits. Curtailment and captured-price analyses further demonstrate that the entropy-based configuration improves both utilization efficiency and economic performance, highlighting its practical value for offshore renewable integration.

Spatial evaluation of the four development areas indicates that Areas 2 and 3 exhibit the most favorable combination of complementarity, stability, and economic

alignment, making them suitable as priority zones for grid-connected multi-energy deployment. Area 4 possesses substantial resource potential but requires flexible loads such as hydrogen production or desalination to absorb surplus generation, while Area 1 is better aligned with small-scale demonstration development.

Overall, the findings confirm that coordinated exploitation of wind, wave, tidal, and solar energy can form a resilient supply system for the Zhoushan Archipelago, reducing storage dependence and mitigating extreme fluctuations. The entropy-weighted approach developed in this study offers a transferable and quantitatively grounded framework for integrated renewable energy planning in archipelagic and coastal regions.

Future research should incorporate high-resolution observational datasets, detailed storage and grid models, and multi-scenario techno-economic optimization to extend the analysis from resource complementarity to full operational planning. Such advancements will support the design of next-generation MECS capable of achieving high renewable penetration in island grids.

Acknowledgements

Appendix A. Calculation Methods for Capacity Factors

Capacity Factor (CF) is a key indicator for measuring the energy output efficiency of power generation facilities. It is defined as the ratio of the system's actual total energy output to its maximum theoretical output capacity at full load operation. In this study, the capacity factor is used to evaluate the resource utilization potential of each renewable energy source in the Zhoushan Archipelago region. The general formula for *CF* is shown in Equation (1):

$$CF = \frac{\overline{P_t}}{P_e} \tag{3}$$

Among them, P_t indicates the average power, which is the ratio of actual energy output to time; P_e symbolizes the installed capacity, which is the maximum power that the power generating facility is planned to supply constantly and stably [38].

Below are the calculation models, parameter settings, and derivation processes for the capacity factors of four energy sources.

A.1. Wind power capacity factor

The wind energy capacity factor can be calculated by following the approach of Wen et al: first, average the instantaneous power Pt over a period of time (e.g., one year), then divide this average by the theoretical annual energy production of the wind turbine operating continuously at its rated power Pe to obtain the capacity factor.

$$CF_{wind} = \frac{P_t}{P_e \times 8760} \tag{4}$$

In equation (2.2), ρ represents the air density, taken as 1.225 kg/m³ in this study; Cp denotes the wind energy utilization coefficient, typically set at 0.4 for modern wind turbines; AT is the swept area of the turbine blades, with a blade diameter of 154 m used in this study; V represents the actual wind speed at the turbine hub height, set at 100 m in this study.

A.2. Wave power capacity factor

The wave energy capacity coefficient can be calculated following the approach by Bahareh Kamranzad et al, using the ratio of the actual annual/monthly power generation of the WEC at a specific site to its theoretical maximum energy output, as shown in Equation (2.4):

$$CF_{wave} = \frac{\sum_{i=1}^{n_T}\sum_{j=1}^{n_H} p_{ij} P_{ij}}{P_e \times 8760} \tag{5}$$

A.3. Tidal power capacity factor

As in Equation (2.5) [32], referencing the fan power curve model, replace “wind speed” with “current flow velocity (v)” and “air density” with “seawater density (ρwater = 1025 kg/m³)”.

$$CF_{tidal} = \frac{P_t}{P_e \times 8760} \tag{6}$$

In the formula, AT represents the swept area of the tidal turbine; P_{tidal} denotes the rated power of the tidal energy converter; Cp is the power coefficient of the turbine (derived from rated parameters, following the same logic as wind energy).

A.4. Solar photovoltaic power capacity factor

Solar CF requires integrating the conversion efficiency of photovoltaic panels with actual solar radiation intensity (accounting for temperature and wind speed effects on

panel performance). First, calculate the actual average power generation capacity, then compare it with the rated power of the photovoltaic panels. Introduce the "performance ratio PR" to correct for environmental factors. As shown in Equation (2.7) [33].

$$CF_{solar} = P_R \frac{RSDS}{RSDS_{STC}} \tag{7}$$

In equation (2.7), RSDS denotes solar radiation density, and STC represents standard test conditions (RSDSSTC = 1000 W/m²).

Appendix B. Detailed formula for entropy rights calculation

Let $X = \{x_{ij}\}_{m\times n}$denote the original CF matrix, where x_{ij}represents the CF of the j-th energy type at hour i. After applying min–max normalization and introducing a small correction constant $\varepsilon = 10^{-6}$ to avoid logarithmic singularities, the standardized matrix $R = \{r_{ij}\}$is obtained. The proportion coefficient for each indicator is then defined as

$$p_{ij} = \frac{r_{ij}}{\sum_i r_{ij}} \tag{8}$$

Based on information theory, the entropy value of the j-th indicator is computed as

$$E_j = -k\sum_{i=1}^{m} p_{ij} \ln p_{ij} \tag{9}$$

where $k = 1/\ln m$. A smaller entropy value reflects larger temporal variability and thus a greater information contribution. The divergence coefficient is defined as $D_j = 1 - E_j$, and the final normalized weight is obtained as

$$\omega_j = \frac{D_j}{\sum_j D_j} \tag{10}$$

Appendix C. Calculation of Curtailment Volume and Capture Costs

Wasted electricity refers to the portion of electricity generated by the system that remains unused due to grid constraints, insufficient transmission capacity, or instantaneous output exceeding local load demand. Its calculation formula is:

$$E_{cur} = \sum_{t=1}^{8784} (\max(0, P_{gen,t} - P_{load,t}) \times \Delta t \tag{11}$$

$P_{gen,t}$ denotes the total power generation of the system at time t, calculated using the formula:

$$P_{gen,t} = \sum_{i} (CF_{i,t} \times P_{Rated,i}) \tag{12}$$

$P_{load,t}$ denotes the actual load demand or grid capacity at time t, serving as the key constraint determining whether curtailment occurs. Based on Zhang et al. (2018)[45], given that the population density of Beiji Island in China is comparable to that of the Zhoushan Archipelago, we adopt a peak load of 800 kW as the reference value. Studies by Husein et al. (2017) [46] and Bertheau et al. (2020) [47] indicate that for small islands dominated by low-energy industries (fisheries, agriculture, etc.), the average load is approximately 0.5 times the peak load. Thus, the average load for the Zhoushan Archipelago can be estimated at 400 kW. Additionally, we have adopted a dynamic load calculation method that incorporates morning and evening peak electricity consumption periods. Therefore, a sine function was selected to generate the dynamic load as an idealized representation; Δt denotes the time step, typically set to 1 hour.

The capture price represents the actual weighted average selling price of renewable energy generation. If P_{cap} exceeds the average market electricity price, it indicates that the system's output aligns well with high-price periods (such as evening peak hours), yielding high economic benefits.

$$P_{cap} = \frac{\sum_{t=1}^{8784} (P_{deliverd,t} \times Price_t \times \Delta t)}{E_{Total}} \tag{13}$$

$P_{delivered,t}$ represents the actual power delivered by the system to the grid or load at time t, calculated as follows:

$$P_{delivered,t} = P_{gen,t} - P_{cur,t} \tag{14}$$

$P_{cur,t}$ represents the curtailment power at time t.

$Price_t$ denotes the electricity market price at time t. According to the 2024 China Power Market Development Report [48], China's average market transaction price is 0.4 yuan per kilowatt-hour. E_{Sold} denotes the system's actual total annual generation.

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